Low-Code Experiment

Exploring Pro-Code Extension of a Zoho Creator Application using Zoho Catalyst / Custom Functions

Sachitha S
241501171
AIML

Abstract
This experiment explores how a low-code Zoho Creator application is extended with code. Custom Deluge functions were created in the Workflow area of the HOTEL BOOKING application, the developer tools of the platform (Application IDE, Widgets, Variables) were reviewed, and an extension using Zoho Catalyst is designed in which a serverless function calculates the bill for a booking and is called from Creator.

1. Aim
To explore pro-code extension of a Zoho Creator application using Zoho Catalyst and custom functions.

Note on Execution
The lab account's free trial has expired (the banner Trial has expired is visible in the screenshots), and the features used in this experiment need a paid plan. The experiment is therefore documented as a procedure together with the evidence that could be captured in the available account. Steps that could not be executed are marked as such.

2. Concepts
Custom functions. Reusable Deluge functions with a name, language, return type and syntax, kept in the Functions tab of the Workflow area and called from workflows, buttons and schedules.

Developer tools. The Settings page lists Developer Tools: Schema Builder, Application IDE, Widgets and Variables. Widgets let developers embed custom HTML, CSS and JavaScript, and the Application IDE gives a code view of the whole application.

Zoho Catalyst. A serverless platform on which functions are written in languages such as Node.js, Java or Python and exposed over HTTPS. A Creator application calls such a function with invokeUrl through a connection, which extends the application beyond what Deluge alone can do.

Use case. A Catalyst function calculates the total bill from the number of nights, the room rate and the tax percentage, and Creator displays or stores the result.

3. Procedure
1. Open the Workflow area and the Functions tab, and create custom Deluge functions in the Default namespace (executed, Figure 1: CountSuites() and Suites()).
2. Open Settings and review the Developer Tools: Schema Builder, Application IDE, Widgets and Variables (executed, Figure 2).
3. Create a Catalyst project and an Advanced I/O function with the Node.js code in section 4, and deploy it (design, not executed).
4. In Creator, create a connection to the Catalyst function and call it with invokeUrl from a form workflow, then store or show the returned total (design, not executed).

4. Deluge / Configuration Code

```
// Zoho Catalyst Advanced I/O function (Node.js): bill calculator
const express = require("express");
const app = express();
app.get("/bill", (req, res) => {
  const nights = parseInt(req.query.nights);
  const rate = parseFloat(req.query.rate);
  const tax = parseFloat(req.query.tax);
  const total = nights * rate * (1 + tax / 100);
  res.json({ nights: nights, total: total });
});
module.exports = app;

// Deluge in Zoho Creator: call the Catalyst function after a booking
resp = invokeUrl
[
    url : "https://<your-catalyst-function-url>/bill?nights=2&rate=3500&tax=12"
    type : GET
    connection : "catalystconn"
];
info resp.get("total");
```

5. Implementation Screenshots (HOTEL BOOKING application)

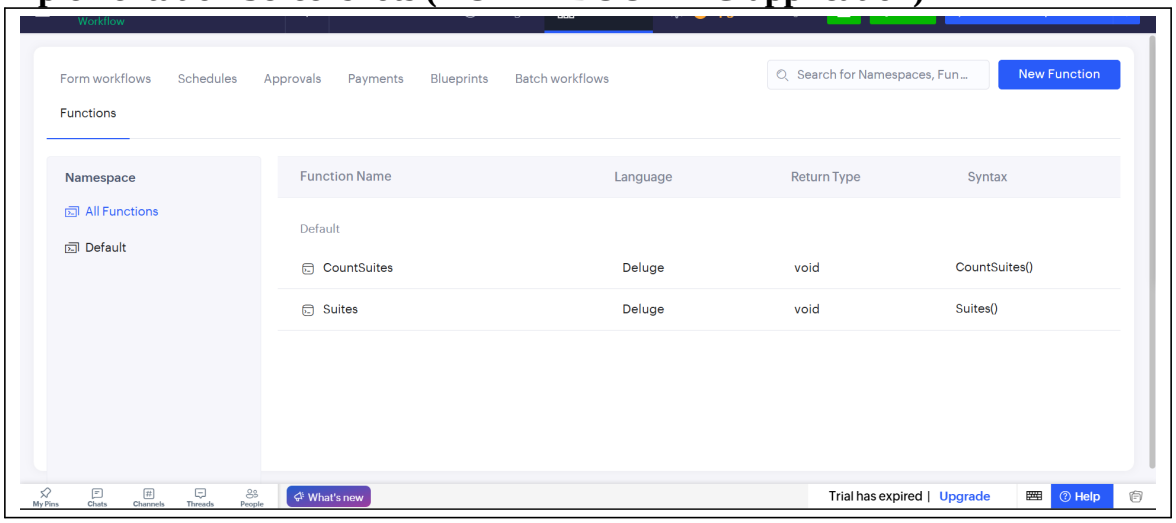


Figure 1: Functions tab of the Workflow area with the custom Deluge functions CountSuites() and Suites()

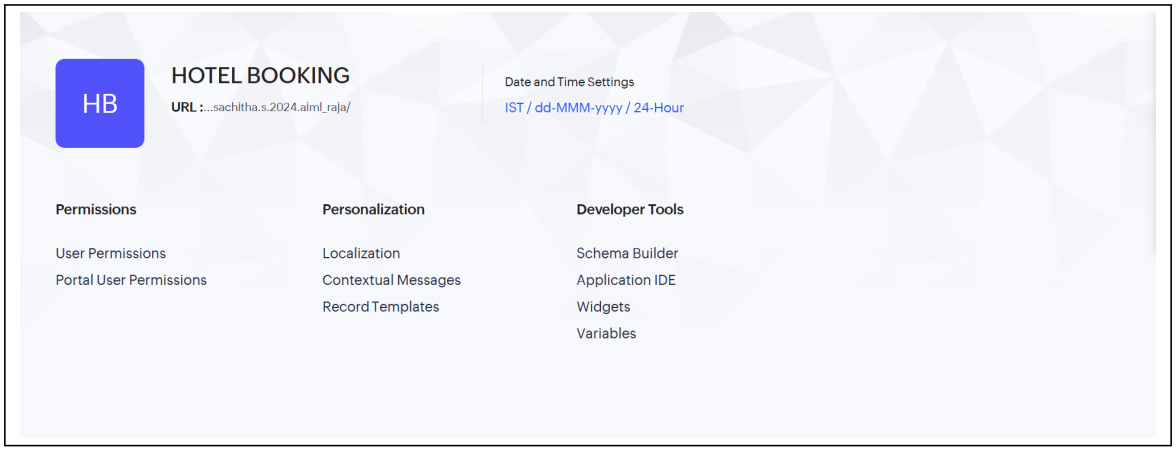


Figure 2: Settings page of HOTEL BOOKING showing the Developer Tools: Schema Builder, Application IDE, Widgets and Variables

6. Expected Output

The Catalyst function returns the total for the given nights, rate and tax as JSON, for example 7840 for 2 nights at 3500 with 12 percent tax, and Creator uses that value in the booking record or confirmation

message.

7. Result

Custom Deluge functions were created in the application and the developer tools were reviewed. The Catalyst function and its call from Creator are given as a design with code, because Catalyst and the connection need a paid plan that the lab account does not have.

8. Learning Outcome

Learnt how custom functions and the developer tools extend a Creator application with code, how a Zoho Catalyst serverless function can be called from Deluge through a connection, and when pro-code extension is worth using over pure low-code.